Analysis of Massive Data Sets

http://www.fer.hr/predmet/avsp

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Analysis of Massive Data Sets: Introduction Lecture

Course Content

- **What are the expected outcomes of the course?**
  - Deal with different types of data
    - High dimensional data
    - Graphs
    - Infinite/never-ending data
    - Labeled data
  - Use different computation models
    - MapReduce
    - Streams and online algorithms
    - In-memory single machine

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Course Content

- **What are the expected outcomes of the course?**
  - Solve real-world problems
    - Recommender Systems
    - Market Basket Analysis
    - Spam filters
    - Duplicate document detection
  - Gain various skills spanning through different areas
    - Linear Algebra (SVD, Rec. Sys., Detecting Communities)
    - Optimization (stochastic gradient descent)
    - Dynamic Programming (Frequent itemsets)
    - Hashing (LSH, Bloom filters)

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Course Content

- **Prerequisites**
  - Algorithms
    - Dynamic programming, basic data structures
  - Basic probability
    - Moments, typical distributions, MLE, …
  - Programming
    - You can do programming assignments in any language we support (Python, Java, C, C++, C# it is your choice)
    - We recommend Python, Java and C#

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Course Content

- **Lectures schedule (1. cycle)**

1. Detecting Near Duplicates – SimHash
2. Detecting Near Duplicates – MinHash
3. Recommendation Systems CF
4. Link Analysis
5. Finding Frequent Itemsets
6. Clustering
7. Communities Detection in SNGs

- Midterm – two weeks break

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Course Content

- **Lectures schedule (2. cycle)**
  8. Mining Data Streams
  9. Mining Dana Streams 2
  10. Recommendation Systems MF
  11. Advertising on the Web
  12. Dimensionality Reduction

  Final exam

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Course Organization

- **Course stuff**
  - Lecturers
    - Associate professor Marin Šilić, PhD
    - Associate professor Goran Delač, PhD
    - Associate professor Klemo Vladimir, PhD
    - Assistant professor Adrian Satja Kurdija, PhD
  - Teaching Assistants
    - Stjepan Požgaj, MSc
  - Questions
    - avsp@fer.hr

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Course Organization

- ECTS points **4**
- Lecture hours **30**
- Programming assignments **15**
- Course duration in weeks **13**

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Course Organization

- Calendar
- Lectures are scheduled
  - Online
  - AVSP YouTube
- Questions
  - avsp@fer.hr

Kalendar nastave za akademsku godinu 2024. / 2025.
Prijediplomski i diplomski studij

	LISTOPAD 2024.					STUDENI 2024.					PROSINAC 2024.					
Tjedan	1	2	3	4	5	6	7	8a	8b		8c	9	10			
Po	30.9	7	14	21	28	4	11	18	25		2	9	16	23	30	
Ut	1	8	15	22	29	5	12	19	26		3	10	17	24	31	
Sr	2	9	16	23	30	6	13	20	27		4	11	18	25		
Če	3	10	17	24	31	7	14	21	28		5	12	19	26		
Pe	4	11	18	25		1	8	15	22	29	6	13	20	27		
Su	5	12	19	26		2	9	16	23	30	7	14	21	28		
Ne	6	13	20	27		3	10	17	24		1	8	15	22	29	

	SIJEČANJ 2025.					VELJAČA 2025.					OŽUJAK 2025.					
Tjedan		11	12	13	14	15a	15b				1	2	3	4	5	
Po		6	13	20	27	3	10	17	24		3	10	17	24	31	
Ut		7	14	21	28	4	11	18	25		4	11	18	25		
Sr	1	8	15	22	29	5	12	19	26		5	12	19	26		
Če	2	9	16	23	30	6	13	20	27		6	13	20	27		
Pe	3	10	17	24	31	7	14	21	28		7	14	21	28		
Su	4	11	18	25		1	8	15	22		1	8	15	22	29	
Ne	5	12	19	26		2	9	16	23		2	9	16	23	30	

	TRAVANJ 2025.					SVIBANJ 2025.					LIPANJ 2025.					
Tjedan	5	6	7	8a	8b	8b	8c	9	10	11		12	13	14	15a	15b
Po		7	14	21	28		5	12	19	26		2	9	16	23	30
Ut	1	8	15	22	29		6	13	20	27		3	10	17	24	
Sr	2	9	16	23	30		7	14	21	28		4	11	18	25	
Če	3	10	17	24		1	8	15	22	29		5	12	19	26	
Pe	4	11	18	25		2	9	16	23	30		6	13	20	27	
Su	5	12	19	26		3	10	17	24	31		7	14	21	28	
Ne	6	13	20	27		4	11	18	25		1	8	15	22	29	

	SRPANJ 2025.					KOLOVOZ 2025.					RUJAN 2025.					
Tjedan	15b															
Po		7	14	21	28		4	11	18	25	1	8	15	22	29	
Ut	1	8	15	22	29		5	12	19	26	2	9	16	23	30	
Sr	2	9	16	23	30		6	13	20	27	3	10	17	24		
Če	3	10	17	24	31		7	14	21	28	4	11	18	25		
Pe	4	11	18	25		1	8	15	22	29	5	12	19	26		
Su	5	12	19	26		2	9	16	23	30	6	13	20	27		
Ne	6	13	20	27		3	10	17	24	31	7	14	21	28		

30.9.	Početak nastave	16	Sjednice Fakultetskog vijeća
1	Nastava	22	Dan Fakulteta
19	Međuispiti	1	Blagdani
10	Ispiti	21	Kolektivni godišnji odmor
		20	Božićni domjenak

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Course Organization

- **Programming assignments**
  - 6 programming assignments
  - Students submit their solutions independently
  - The solutions will be evaluated using online judge system
  - Follow the news on the course website, instructions for the first assignment will be posted soon!

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Course Organization

Grades

- **Theory** — **70 points**
  - Midterm — 30 points
  - Final exam — 30 points
  - Lectures attendance — 10 points
- **Programming assignments** — **30 points**
  - Each assignment will be graded uniformly

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Course Organization

- **The right to take an exam**
  - Programming assignments threshold — 15 points (50%)
- **To get a positive grade**
  - Theory threshold — 35 points (50%)
  - Programming assignments threshold — 15 points (50%)

- **Grades** (*)
  - excellent (5) — ≥ 88 points
  - very good (4) — ≥ 75 points
  - good (3) — ≥ 63 points
  - sufficient (2) — ≥ 50 points

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